

AP-E3 Nonlinear Follow-up: A Charged Two-Colour EFT Proxy, Diquark Gap, and the Necessary $B = 1$ Soliton-Hessian Gate

Route F research note

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Abstract

We construct and solve a deliberately limited low-energy proxy for the unsettled strong-phase and soliton-stability gates of the charged two-colour AP-E3 candidate. A real $O(4)$ meson multiplet is supplemented by a complex $U(1)_g$ -charge-two diquark. Its homogeneous potential yields an analytic phase boundary, vacuum condensates, and the exact tree-level gaps

$$m_\pi^2 = \frac{h}{f}, \quad m_\sigma^2 = m_\pi^2 + 2\lambda f^2, \quad m_\Delta^2 = m_\pi^2 + \mu_g^2.$$

In the mesonic phase we take the nonlinear $SU(2)$ limit, add the Skyrme term, solve the full $B = 1$ hedgehog boundary-value problem, verify baryon number and Derrick’s virial identity, and diagonalise two quadratic blocks: the generalized spherically symmetric meson Hessian and the most dangerous charged-diquark channel. Grid and volume sequences converge; an intentionally over-attractive diquark coupling produces a negative eigenvalue and therefore tests that the audit can fail.

The term nonperturbative is used only in the narrow classical sense that no small-amplitude or small-profile expansion is used in solving the nonlinear boundary-value problem. This is not lattice two-colour QCD, does not integrate the microscopic gauge theory, and does not close the full three-dimensional coupled Hessian. Moreover, the product target used here excludes the global S^5 escape direction by assumption. The result is consequently a useful necessary gate and a reproducible falsifier, not a proof of the microscopic charged phase or of a stable quantum baryon.

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1 Epistemic boundary and questions answered

The anomaly-consistent charged two-colour construction fixes ultraviolet quantum numbers but not the infrared vacuum. Gauge-anomaly cancellation does not prove a mesonic condensate, a positive diquark gap, or a stable $B = 1$ saddle. Effective descriptions of two-colour dynamics explicitly retain meson and diquark directions [1, 2]; the pseudoreal theory is therefore not equivalent to ordinary three-colour QCD. This note asks three narrower questions.

- (Q1) Can one exhibit a charged effective potential with a transparent meson–diquark phase boundary and a positive benchmark Hessian?
- (Q2) Does its declared mesonic nonlinear subsector possess a numerically converged $B = 1$ hedgehog satisfying the exact topology and scale tests?
- (Q3) Are its radial meson and charged-diquark quadratic blocks locally nonnegative, and can the same implementation detect an unstable control?

The answers are yes inside the proxy. The following statements remain false: a lattice or continuum nonperturbative solution of the microscopic $SU(2)_c \times U(1)_g$ gauge theory; a UV derivation of every effective coefficient; a full three-dimensional Hessian; a fermion-determinant audit; and exclusion of finite-amplitude unwinding paths.

Fail-closed gate 1.1 (Promotion boundary). A positive result below sets only the nonlinear-EFT and radial-Hessian necessary gates to true. It leaves the full three-dimensional Hessian, lattice QC2D, global unwinding, and physics-promotion gates false.

2 Charged homogeneous EFT proxy

2.1 Fields, charge, and potential

Let

$$\varphi = (\sigma, \pi_1, \pi_2, \pi_3, d_1, d_2), \quad \Delta = \frac{d_1 + id_2}{\sqrt{2}}.$$

The mesons are $U(1)_g$ -neutral and Δ has charge $q_\Delta = 2$, as expected for a two-quark operator. Thus

$$D_\mu \Delta = (\partial_\mu - 2ie_g b_\mu) \Delta.$$

With canonical real-field normalization, the proxy Lagrangian is

$$\mathcal{L}_{\text{hom}} = \frac{1}{2}(\partial_\mu \sigma)^2 + \frac{1}{2}(\partial_\mu \boldsymbol{\pi})^2 + |D_\mu \Delta|^2 - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - V, \quad (1)$$

$$V = \frac{\lambda}{4} (\sigma^2 + \boldsymbol{\pi}^2 + 2|\Delta|^2 - v^2)^2 - h\sigma + \mu_g^2 |\Delta|^2. \quad (2)$$

Here μ_g^2 is a matched Pauli–Gürsey-breaking coefficient. Its sign and magnitude are not inferred merely from the gauge charge. Equation (2) is a diagnostic model, not a claimed exact chiral Lagrangian of the UV candidate.

2.2 Mesonic stationary point and exact vacuum Hessian

At $\pi = \Delta = 0$, choose the positive root f of

$$\lambda(f^2 - v^2)f = h. \quad (3)$$

Define $m_\pi^2 = h/f = \lambda(f^2 - v^2)$. Direct differentiation of Eq. (2) gives

$$\nabla^2 V|_{(f,0,0)} = \text{diag}(m_\sigma^2, m_\pi^2, m_\pi^2, m_\pi^2, m_\Delta^2, m_\Delta^2), \quad (4)$$

where

$$m_\sigma^2 = m_\pi^2 + 2\lambda f^2, \quad m_\Delta^2 = m_\pi^2 + \mu_g^2. \quad (5)$$

Therefore the charged direction is locally stable precisely when

$$\mu_g^2 > -m_\pi^2. \quad (6)$$

The two identical diquark eigenvalues follow from unbroken $U(1)_g$, and linear meson–diquark mixing is forbidden by charge.

2.3 Condensed branch and continuous onset

When $\Delta \neq 0$, the stationary equations read

$$\lambda(R^2 - v^2)\sigma = h, \quad [\lambda(R^2 - v^2) + \mu_g^2]d_a = 0, \quad R^2 = \sigma^2 + d_1^2 + d_2^2. \quad (7)$$

For $\mu_g^2 < 0$, the charged branch is consequently

$$\sigma = -\frac{h}{\mu_g^2}, \quad d_1^2 + d_2^2 = v^2 - \frac{\mu_g^2}{\lambda} - \frac{h^2}{(\mu_g^2)^2}. \quad (8)$$

The right-hand side vanishes continuously at $\mu_g^2 = -h/f = -m_\pi^2$. Below that point $U(1)_g$ is Higgsed and the homogeneous Hessian contains the corresponding angular zero mode before gauge fixing. This analytic onset is used as a positive and negative classification test in the scan.

2.4 Explicit $(e_g, v_g/\Lambda_c)$ sensitivity chart

The EFT coefficient μ_g^2 is not yet known from the charged microscopic theory. To avoid hiding that ignorance while still scanning the requested coordinates, we declare the matching ansatz

$$\frac{\mu_g^2}{\Lambda_c^2} = \frac{\mu_{\text{PG},0}^2}{\Lambda_c^2} + c_g \left(e_g \frac{v_g}{\Lambda_c} \right)^2. \quad (9)$$

The numerical chart uses $\mu_{\text{PG},0}^2/\Lambda_c^2 = -0.40$, $c_g = 0.24$, and $\Lambda_c = f = 1$. This makes both sides of the analytic phase boundary visible. It is a sensitivity coordinate chart, not a loop matching result. Every cell records e_g , v_g/Λ_c , $\langle \sigma \rangle$, $|\langle d \rangle|$, and the smallest homogeneous Hessian eigenvalue. Table 1 shows the complete deterministic chart. Here D denotes the diquark-condensed branch and M the mesonic branch. The zero on D is the ungauged-fixed $U(1)_g$ angular mode, not a negative curvature.

Table 1: Declared charged-coordinate sensitivity chart.

e_g	v_g/Λ_c	μ_g^2/Λ_c^2	phase	σ/Λ_c	$(d , \lambda_{\min})$
0.00	0.50	-0.4000	D	0.625000	(0.796477,0)
0.00	1.00	-0.4000	D	0.625000	(0.796477,0)
0.00	2.00	-0.4000	D	0.625000	(0.796477,0)
0.00	3.00	-0.4000	D	0.625000	(0.796477,0)
0.25	0.50	-0.3963	D	0.630915	(0.791405,0)
0.25	1.00	-0.3850	D	0.649351	(0.775141,0)
0.25	2.00	-0.3400	D	0.735294	(0.688725,0)
0.25	3.00	-0.2650	D	0.943396	(0.335416,0)
0.50	0.50	-0.3850	D	0.649351	(0.775141,0)
0.50	1.00	-0.3400	D	0.735294	(0.688725,0)
0.50	2.00	-0.1600	M	1.000000	(0,0.090000)
0.50	3.00	0.1400	M	1.000000	(0,0.250000)
0.75	0.50	-0.3663	D	0.682594	(0.743936,0)
0.75	1.00	-0.2650	D	0.943396	(0.335416,0)
0.75	2.00	0.1400	M	1.000000	(0,0.250000)
0.75	3.00	0.8150	M	1.000000	(0,0.250000)
1.00	0.50	-0.3400	D	0.735294	(0.688725,0)
1.00	1.00	-0.1600	M	1.000000	(0,0.090000)
1.00	2.00	0.5600	M	1.000000	(0,0.250000)
1.00	3.00	1.7600	M	1.000000	(0,0.250000)

3 Mesonic nonlinear limit and the $B = 1$ saddle

3.1 Why a product-target proxy is used

In the positive-gap phase, freeze the heavy radial mode and set $\Delta = 0$. Write

$$U = \frac{\sigma + i\boldsymbol{\tau} \cdot \boldsymbol{\pi}}{f} \in SU(2). \quad (10)$$

We then use the standard sigma plus Skyrme energy [3, 4]. This is a product-target construction: U remains an $SU(2)$ coordinate when the spectator Δ fluctuates. It therefore retains $\pi_3(SU(2)) = \mathbb{Z}$.

That choice is useful for a local gate but logically costly. In ungauged two-colour QCD the Pauli–Gürsey target for two flavours is $SU(4)/Sp(4) \simeq S^5$, for which $\pi_3(S^5) = 0$. The present proxy does not prove that the charged microscopic theory dynamically factorizes its target, nor does it exclude a finite-amplitude path that restores an S^5 -like escape direction. This is why local Hessian positivity is not promoted to global soliton stability.

3.2 Static energy and hedgehog reduction

Take

$$E_U = \int d^3x \left[\frac{f^2}{4} \text{Tr}(\partial_i U \partial_i U^\dagger) + \frac{1}{32e^2} \text{Tr}([L_i, L_j]^\dagger [L_i, L_j]) + \frac{f^2 m_\pi^2}{2} \text{Tr}(1 - U) \right], \quad (11)$$

where $L_i = U^\dagger \partial_i U$. For

$$U(\mathbf{x}) = \cos F(r) + i\boldsymbol{\tau} \cdot \hat{\mathbf{x}} \sin F(r), \quad x = efr, \quad \beta = \frac{m_\pi}{ef}, \quad (12)$$

the energy is

$$E_U = \frac{4\pi f}{e} \int_0^\infty dx \mathcal{E}_F, \quad (13)$$

$$\mathcal{E}_F = \frac{1}{2}x^2 F'^2 + \sin^2 F(1 + F'^2) + \frac{\sin^4 F}{2x^2} + \beta^2 x^2(1 - \cos F). \quad (14)$$

The baryon number reduces to

$$B = -\frac{2}{\pi} \int_0^\infty \sin^2 F F' dx = \frac{1}{\pi} \left[F - \frac{1}{2} \sin 2F \right]_{x=\infty}^{x=0}. \quad (15)$$

Thus $F(0) = \pi$, $F(\infty) = 0$ imply $B = 1$.

3.3 Euler–Lagrange equation and regular origin data

Varying Eq. (14) yields

$$(x^2 + 2 \sin^2 F)F'' + 2xF' + 2 \sin F \cos F(F'^2 - 1) - \frac{2 \sin^3 F \cos F}{x^2} - \beta^2 x^2 \sin F = 0. \quad (16)$$

A numerically stable origin condition cannot set $F(\epsilon) = \pi$ while allowing a finite nonzero slope. Substitution of

$$F(x) = \pi - ax + bx^3 + O(x^5) \quad (17)$$

into Eq. (16) gives

$$b = \frac{a(2a^4 + 4a^2 + 3\beta^2)}{30(1 + 2a^2)}. \quad (18)$$

The boundary-value solver treats a as an unknown parameter and imposes both Eq. (18) at $x = \epsilon$ and $F(L) = 0$.

3.4 Derrick identity and collapse negative control

Split the dimensionless integral into $E_2 + E_4 + E_0$, corresponding to the two-derivative, four-derivative, and mass terms. Under $F(x) \mapsto F(x/\lambda)$,

$$E(\lambda) = \lambda E_2 + \lambda^{-1} E_4 + \lambda^3 E_0. \quad (19)$$

Any finite-size stationary solution must obey

$$E_2 - E_4 + 3E_0 = 0, \quad E''(1) = 2E_4 + 6E_0 > 0. \quad (20)$$

Deleting the Skyrme contribution leaves $E'_{\text{no Skyrme}}(1) = E_2 + 3E_0 > 0$, so shrinking lowers the energy. The code checks both the virial identity and this deliberate collapse control.

4 Second variation and spectral problems

4.1 Spherically symmetric meson block

Let $F \mapsto F + \varepsilon \eta$ with $\eta(0) = \eta(L) = 0$. Write

$$\mathcal{E}_F = \frac{1}{2}A(F, x)F'^2 + \mathcal{U}(F, x), \quad A = x^2 + 2 \sin^2 F, \quad (21)$$

$$\mathcal{U} = \sin^2 F + \frac{\sin^4 F}{2x^2} + \beta^2 x^2(1 - \cos F). \quad (22)$$

Twice differentiating the energy and integrating the mixed $\eta\eta'$ term by parts gives

$$\delta^2 E_F = \frac{4\pi f}{e} \int_0^L dx [A\eta'^2 + V_F \eta^2], \quad (23)$$

$$V_F = \mathcal{U}_{FF} + \frac{1}{2} A_{FF} F'^2 - (A_F F')', \quad (24)$$

or, using the background equation,

$$\begin{aligned} V_F &= 2 \cos 2F + \frac{2}{x^2} (3 \sin^2 F \cos^2 F - \sin^4 F) + \beta^2 x^2 \cos F \\ &\quad - 2 \cos 2F F'^2 - 4 \sin F \cos F F''. \end{aligned} \quad (25)$$

The same sigma and Skyrme terms give the radial kinetic norm $\int A \eta^2 dx$. Hence the displayed squared frequencies solve the generalized Sturm–Liouville problem

$$\mathcal{H}_F \eta = \omega^2 A \eta, \quad \mathcal{H}_F = -\frac{d}{dx} A \frac{d}{dx} + V_F. \quad (26)$$

At large x , the continuum floor is $\omega^2 = \beta^2$. A lowest finite-box eigenvalue approaching β^2 from above is therefore not a localized breathing mode.

4.2 Charged-diquark block

The leading symmetry-allowed static spectator energy is chosen as

$$E_{\Delta}^{(2)} = \frac{4\pi f}{e} \sum_{a=1}^2 \frac{1}{2} \int_0^\infty dx x^2 [(d'_a)^2 + \{\alpha^2 + \chi(1 - \cos F)\} d_a^2], \quad (27)$$

where

$$\alpha = \frac{m_{\Delta}}{ef}. \quad (28)$$

The coefficient χ measures the matched core interaction. It is independent input and is scanned with a tachyonic negative control. Because the background has $\Delta = 0$, the two real components are degenerate and there is no quadratic meson–diquark mixing.

For angular momentum ℓ , set $u_{\ell} = x d_{\ell}$. The Hessian becomes

$$\mathcal{H}_{\Delta, \ell} = -\frac{d^2}{dx^2} + \alpha^2 + \chi(1 - \cos F) + \frac{\ell(\ell+1)}{x^2}, \quad u_{\ell}(0) = u_{\ell}(L) = 0. \quad (29)$$

Thus $\ell = 0$ is the most dangerous scalar channel: $\mathcal{H}_{\Delta, \ell} - \mathcal{H}_{\Delta, 0} \geq 0$. Positivity of its lowest eigenvalue closes the charged-scalar block in this proxy. It does *not* close nonradial meson, vector, gauge, radial- σ , ghost, or fermionic channels. A positive bound mode satisfies

$$0 < \omega_{\Delta, 0}^2 < \alpha^2. \quad (30)$$

4.3 Gauge-field block at the declared saddle

At $\Delta = 0$, the Maxwell field does not mix linearly with d_1, d_2 . With nonnegative Maxwell and optional Proca/Higgs quadratic terms, its isolated tree-level block is nonnegative. This observation does not replace gauge fixing and ghost accounting in the full coupled three-dimensional problem; those remain explicit blockers.

5 Deterministic numerical experiment

5.1 Benchmark and algorithms

The benchmark fixes

$$f = e = 1, \quad \lambda = 6, \quad m_{\pi} = 0.5, \quad \mu_g^2 = 0.56, \quad \chi = -2, \quad (31)$$

so that

$$v^2 = \frac{23}{24}, \quad h = \frac{1}{4}, \quad (m_\pi, m_\sigma, m_\Delta) = (0.5, 3.5, 0.9), \quad (\beta, \alpha) = (0.5, 0.9).$$

The BVP is solved on $[\epsilon, L]$, $\epsilon = 10^{-4}$, using the regular series and an adaptive collocation tolerance of 3×10^{-7} . The radial generalized problem uses linear finite elements with consistent mass matrix. The transformed diquark problem uses a centered second-difference operator. Sparse symmetric eigensolvers request the algebraically lowest eigenvalues.

5.2 Benchmark results

The machine-readable output is authoritative for the final digits. The expected audit structure is

Quantity	Result	Interpretation
Vacuum gaps	(0.5, 3.5, 0.9)	positive meson, radial, diquark gaps
B and $ B - 1 $	0.99999999594, 4.06×10^{-10}	topology/integration check
$E_2 - E_4 + 3E_0$	-1.55×10^{-7}	relative residual 2.52×10^{-8}
lowest ω_F^2	0.311400567	$L = 18, N = 1200$ radial gate
lowest $\omega_{\Delta,0}^2$	0.696506101	positive bound mode below $\alpha^2 = 0.81$
$\chi = -5$ control	-1.430515810	charged tachyon detected

Three grids with $N = 600, 1200, 2400$ at fixed $L = 18$ test discretization. A second sequence with $(L, N) = (14, 700), (18, 900), (22, 1100)$ tests the box. The soliton energy and localized diquark eigenvalue must stabilize. The first radial finite-box state is instead expected to decrease toward the continuum floor $\beta^2 = 0.25$, remaining positive.

N	spacing	$\omega_{F,\min}^2$	$\omega_{\Delta,\min}^2$
600	0.02999983	0.311401592	0.696441268
1200	0.01499992	0.311400567	0.696506101
2400	0.00749996	0.311400310	0.696522301

Table 3: Fixed-box discretization sequence. Both observed convergence orders are approximately two; the machine-readable card records Richardson extrapolates and error estimates.

L	N	$\int \mathcal{E}_F dx$	$\omega_{F,\min}^2$	$\omega_{\Delta,\min}^2$
14	700	6.135914701	0.349739800	0.696547693
18	900	6.135914193	0.311400833	0.696489297
22	1100	6.135914184	0.291384645	0.696485381

Table 4: Fixed-spacing volume sequence. The radial finite-box state moves toward $\beta^2 = 0.25$, whereas the localized charged mode stabilizes.

5.3 What the negative controls prove

Three failure modes are built into the same calculation.

1. Setting $\mu_g^2 < -m_\pi^2$ makes the mesonic stationary point unstable and moves the homogeneous minimum to the charged branch.
2. Removing E_4 violates the possibility of a finite-size Derrick stationary point and produces collapse.
3. Replacing the benchmark $\chi = -2$ by $\chi = -5$ makes the lowest charged-diquark eigenvalue negative.

Consequently a green benchmark is not caused by an eigenvalue routine that silently takes absolute values or by a phase classifier that always selects the mesonic branch.

6 Interpretation and remaining routes

6.1 What has been learned

The proxy demonstrates a logically consistent local scenario:

$$\begin{aligned} m_\Delta^2 > 0, \quad \langle \Delta \rangle = 0, \quad U(1)_g \text{ unbroken,} \\ B = 1, \quad \delta^2 E_F > 0 \text{ in the radial sample,} \quad \delta^2 E_\Delta > 0 \text{ for every scalar } \ell. \end{aligned} \tag{32}$$

It also quantifies how a sufficiently attractive soliton core can destroy that scenario even while the homogeneous vacuum retains a positive diquark gap. Vacuum stability and soliton-core stability are therefore independent gates.

6.2 What has not been learned

The following gaps cannot be repaired by increasing the radial grid.

1. The ansatz (9) and the coupling χ need matching from the microscopic charged gauge theory.
2. The mesonic $SU(2)$ product factor is assumed. A faithful full target may allow global unwinding even when every local eigenvalue reported here is positive.
3. Nonradial meson deformations, the finite radial- σ field, vector mesons, gauge/ghost fields, and collective zero modes require a coupled three-dimensional Hessian.
4. Fermionic and gauge determinants can shift the classical spectrum.
5. A lattice calculation must measure the meson and diquark condensates, charged gap, baryon free energy, and finite-volume scaling.

6.3 Recommended next implementations

There are three useful continuations, ordered by evidential strength.

1. **Lattice phase chart.** Scan $(e_g, v_g/\Lambda_c)$ and measure $\langle \bar{q}q \rangle$, a gauge-invariant dressed diquark correlator, the corresponding gap, and volume scaling. This directly replaces the matching ansatz.
2. **Full EFT Hessian.** Retain σ , vector mesons and the massive $U(1)_g$ field on a three-dimensional grid, gauge-fix, remove collective modes, and continue the lowest eigenpairs in χ .
3. **Global path search.** Use a string or nudged-elastic-band method in the faithful field space between the $B = 1$ configuration and vacuum. This tests finite-amplitude escape that a Hessian cannot see.

7 Reproducibility ledger

Run from the repository root:

```
python3 route_f/code/scan_ap_e3_charged_two_colour_proxy.py
```

The program writes

```
route_f/output/ap_e3_charged_two_colour_proxy.json  
route_f/output/ap_e3_charged_two_colour_proxy.md
```

and records Python, NumPy, SciPy, source hashes, every mechanical assertion, the phase chart, convergence sequences, Hessian spectra, negative controls, and fail-closed booleans.

References

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